PALLAVI KWATRA

Data Engineer | GCP Cloud Data Engineer
System Engineer | Data Platform Engineer
AI & Cloud Platform Engineer

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SUMMARY

- **Google Cloud Certified** Data Engineer with **5+** years of experience **architecting** and **scaling** enterprise-grade data platforms on **GCP** within highly regulated banking and cybersecurity environments.
- **Designed** high-volume batch and streaming **ETL** pipelines processing **20M**+ daily records using **BigQuery**, **Dataflow** (Apache Beam), Pub/Sub, and **Cloud Composer** (**Airflow**).
- Delivered **GDPR**-compliant cloud-native architectures, reducing **audit** time by 90% and improving regulatory visibility across **multi-cloud** environments.
- Strong expertise in Infrastructure-as-Code (**Terraform**), **Kubernetes**, **CI/CD** automation, and cost-optimised **cloud architecture**.
  Google **Generative AI Leader certified**, applying AI-driven automation and governance strategies to modern data platforms.

WORK HISTORY

Lead Data Engineer - Cloud Security

Deutsche Bank (Tata Consultancy Services) **Jan. 2025 to Current**

- **Architected** and deployed a cloud-native security analytics and governance platform using Terraform, BigQuery, Cloud Composer (**Airflow**), and **Looker**, enabling automated **compliance** monitoring across **5,000+** cloud assets and reducing manual regulatory effort by **90%.**
- Designed high-volume **ingestion** frameworks processing **8TB+** monthly audit logs via Cloud Logging, **LogSink**, and **Pub/Sub** into **BigQuery**, improving real-time **threat detection** and reducing security incident response time by **65%.**
- Implemented **VPC** Service Controls (VPC-SC) to enforce data perimeter **security** for sensitive banking workloads, preventing unauthorised **data exfiltration** and strengthening **GDPR** and **FCA** compliance across restricted service boundaries.
- Led end-to-end delivery and security validation of **Akamai**-**WAF** integrations, analysing **DDoS** attack patterns and domain-level vulnerabilities using **Cloud Logging telemetry**; improved attack visibility by **70%** and enhanced real-time mitigation readiness for critical financial applications.
- Leading a **Generative AI** Proof-of-Concept to develop an AI-powered **chatbot** integrated with enterprise **GCP** datasets to automate **cybersecurity** audit reporting and build an AI-driven **report writer**, projected to reduce audit documentation turnaround time by **50%** while improving reporting accuracy and standardisation.

Data Engineer – PAM Inventory & Compliance Automation

Deutsche Bank (Tata Consultancy Services) **July 2023 to Jan. 2025**

- Built high-throughput **ETL** pipelines using **Dataflow** (Apache Beam - **Python**) ingesting **20M+** daily records from **APIs** and on-prem databases into BigQuery, reducing audit report turnaround time by **30%.**
- Automated **end-to-end** compliance workflows integrating IAM, **GCS**, **Cloud** Scheduler, Secret Manager, and **Firewall**, improving privileged access monitoring coverage by **45%.**
- Identified and remediated **700+** misconfigured privileged accounts, closing 90% of audit gaps that persisted for 8+ years and mitigating potential regulatory penalties.
- Improved BigQuery **cost efficiency** by **25**% through query optimisation, table partitioning, and workload tuning, saving approximately **£95K** annually.

Full Stack Developer

Credit Suisse Bank (Tata Consultancy Services) **Feb.2023 to July 2023**

- Developed and **containerised** microservices using **Java**, **Spring Boot**, **Docker**, and **OpenShift**, reducing deployment time by **70**% via **CI/CD** automation (**Jenkins**).
- **Migrated** legacy monolithic systems to microservices architecture, increasing **scalability** by 40% and reducing production incidents by **35**%.
- Enhanced **data validation** and routing **workflows**, improving transaction processing efficiency by **30**% across financial request **pipelines**.
- Implemented **logging** and monitoring enhancements, reducing mean time to resolution (**MTTR**) by **25**%.

Software Developer

Credit Suisse Bank (Tata Consultancy Services) **Nov. 2020 to Feb. 2023**

- Engineered enterprise-grade **loan processing** systems using Java, Spring Boot, **REST APIs**, and **PL/SQL**, supporting **500K**+ monthly transactions.
- Optimised backend data **transformation pipelines** using parallel processing, reducing **response latency** by **30**% and improving system throughput by 25%.
- Migrated **monolithic application** to microservices, increasing system reliability by **40**% and improving deployment stability.
- Improved **database indexing** and **query performance**, reducing critical batch job runtimes from 4 hours to under 2.5 hours.

EDUCATION

Bachelor of Computer Science Engineering

Dr. K.N. Modi Institute of Engineering & Technology **June 2016 to July 2020**

SKILLS

CLOUD & DATA ENGINEERING	Google Cloud Platform (GCP), BigQuery, Dataflow, Apache Beam, Pub/Sub, Cloud Composer (Airflow), GCS
ETL & BIG DATA	Batch & Streaming Pipelines, ELT/ETL Design, Hadoop, Hive, MapReduce, Data Warehousing, Data Modelling
DEVOPS & ARCHITECTURE	Terraform (Infrastructure-as-Code), Kubernetes, Docker, OpenShift, CI/CD, Jenkins
DATA VISUALIZATION	Tableau, Power BI, Looker
TOOLS	GIT, IntelliJ, Visual Studio, Swagger, Postman, WinSCP, Bruno
PROGRAMMING & DATABASES	Python, Java, SQL, PL/SQL, Bash, Spring Boot, REST APIs
DATA GOVERNANCE & SECURITY	GDPR Compliance, Audit Automation, Access Controls, Policy Enforcement, Cybersecurity Analytics

CERTIFICATIONS

Google Generative AI Leader Certification – Google Cloud (2025)
Google Cloud Certified – Professional Cloud Security Engineer (2025)
Looker Development Certification (2024)
Google Cloud Certified – Professional Cloud Architect (2024)
Google Cloud Certified – Associate Cloud Engineer (2024)